

Notes on 0PN, 1PN, and TaylorF2 Chirp Parametrisations

1 TaylorF2 phase convention

The TaylorF2 waveform is a frequency-domain stationary-phase approximation. For a nonspinning binary, its phase can be written schematically as

$$\Psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{128\eta} v^{-5} \sum_{k=0}^7 [a_k + b_k \log v] v^k, \quad (1)$$

where

$$v = (\pi M f)^{1/3}, \quad M = m_1 + m_2, \quad \eta = \frac{m_1 m_2}{(m_1 + m_2)^2}. \quad (2)$$

The index k corresponds to PN order $k/2$. Thus terms through $k = 7$ correspond to phase corrections through 3.5PN. In the LAL TaylorF2 call used here, with no explicit phase-order override and zero spins, the default nonspinning TaylorF2 phasing includes nonzero terms up to 3.5PN.

2 The 1PN TaylorF2 phase term

Keeping only the Newtonian and 1PN TaylorF2 phase terms gives

$$\Psi_{0\text{PN}+1\text{PN}}(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{128\eta} v^{-5} [1 + \alpha_2 v^2], \quad (3)$$

with

$$\alpha_2 = \frac{3715}{756} + \frac{55}{9}\eta. \quad (4)$$

For an equal-mass binary, $\eta = 1/4$, so

$$\alpha_2 = \frac{3715}{756} + \frac{55}{36}. \quad (5)$$

3 Factoring out frequency

Up to the constant phase offset $-\phi_c - \pi/4$, the 0PN+1PN phase can be written as

$$\Psi_{0\text{PN}+1\text{PN}}(f) = \text{const} + 2\pi f t_c + A f^{-5/3} + B f^{-1}, \quad (6)$$

where

$$A = \frac{3}{128\eta} (\pi M)^{-5/3}, \quad (7)$$

$$B = \frac{3}{128\eta} \alpha_2 (\pi M)^{-1}. \quad (8)$$

Therefore one may factor out $2\pi f$ and define an effective frequency-dependent time coordinate,

$$\Psi_{0\text{PN}+1\text{PN}}(f) = \text{const} + 2\pi f \tau_{\text{F2}}(f), \quad (9)$$

where

$$\tau_{\text{F2}}(f) = t_c + \frac{A}{2\pi} f^{-8/3} + \frac{B}{2\pi} f^{-2}. \quad (10)$$

The same algebraic factorisation is possible through 3.5PN. In general,

$$\Psi(f) = \text{const} + 2\pi f \tau_{\text{F2}}(f), \quad (11)$$

with

$$\tau_{\text{F2}}(f) = t_c + \frac{3}{256\pi\eta} \sum_{k=0}^7 \left[a_k + b_k \log \left((\pi M f)^{1/3} \right) \right] (\pi M)^{(k-5)/3} f^{(k-8)/3}. \quad (12)$$

The corresponding powers of f inside τ_{F2} are

PN order	k	power in $\tau_{\text{F2}}(f)$
0PN	0	$f^{-8/3}$
0.5PN	1	$f^{-7/3}$
1PN	2	f^{-2}
1.5PN	3	$f^{-5/3}$
2PN	4	$f^{-4/3}$
2.5PN	5	f^{-1}
3PN	6	$f^{-2/3}$
3.5PN	7	$f^{-1/3}$

Some PN orders also include logarithmic corrections.

This factorisation is an algebraic statement about the phase. For the physical chirp trajectory, the relevant time-frequency relation is instead

$$t(f) = \frac{1}{2\pi} \frac{d\Psi}{df}. \quad (13)$$

This derivative relation is what should be used when extracting a chirp trajectory from TaylorF2.

4 0PN beta parametrisation

At Newtonian order, the chirp can be written explicitly as

$$f(\beta, t) = f_0 \left(1 - \frac{8}{3} \beta t \right)^{-3/8}. \quad (14)$$

Here f_0 is the frequency at reference time $t = 0$, and

$$\beta = \frac{96}{5} \pi^{8/3} \left(\frac{GM_c}{c^3} \right)^{5/3} f_0^{8/3}, \quad (15)$$

where

$$M_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \quad (16)$$

is the chirp mass. The corresponding phase evolution is

$$\phi(\beta, t) = -\frac{6\pi}{5} f_0 \frac{\left(1 - \frac{8}{3} \beta t \right)^{5/8}}{\beta} + \phi_0. \quad (17)$$

An additive constant may be chosen so that $\phi(0) = \phi_0$ if desired.

5 Can the same construction be done at 1PN?

At 1PN, the frequency evolution is still separable, but it is no longer described by the single clean 0PN beta form. The 1PN evolution may be written as

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_c}{c^3} \right)^{5/3} f^{11/3} \left[1 - \alpha \left(\pi \frac{GM}{c^3} f \right)^{2/3} \right], \quad (18)$$

where

$$\alpha = \frac{743}{336} + \frac{11}{4} \eta. \quad (19)$$

For equal masses, $\eta = 1/4$, giving $\alpha = 487/168$.

Using the same β as in Eq. (15), define

$$x = \frac{f}{f_0}, \quad \epsilon = \alpha \left(\pi \frac{GM}{c^3} f_0 \right)^{2/3}. \quad (20)$$

Then Eq. (18) becomes

$$\frac{df}{dt} = \beta f_0 x^{11/3} \left[1 - \epsilon x^{2/3} \right]. \quad (21)$$

For a generic mass ratio, ϵ is an additional independent parameter. However, for the equal-mass case used in the numerical examples, $\eta = 1/4$ and

$$M = 2^{6/5} M_c. \quad (22)$$

Therefore ϵ can be eliminated in favour of β and f_0 . From Eq. (15),

$$\left(\frac{GM_c}{c^3} \right)^{5/3} = \frac{5\beta}{96\pi^{8/3} f_0^{8/3}}, \quad (23)$$

so that

$$\pi \frac{GM}{c^3} f_0 = \left(\frac{5\beta}{24\pi f_0} \right)^{3/5}. \quad (24)$$

For equal masses, $\alpha = 487/168$, and hence

$$\epsilon(\beta, f_0) = \frac{487}{168} \left(\frac{5\beta}{24\pi f_0} \right)^{2/5}. \quad (25)$$

The equal-mass 1PN chirp can therefore be written as

$$\frac{df}{dt} = \beta f_0 x^{11/3} \left[1 - \frac{487}{168} \left(\frac{5\beta}{24\pi f_0} \right)^{2/5} x^{2/3} \right], \quad x = \frac{f}{f_0}. \quad (26)$$

Thus, if f_0 is fixed, the equal-mass 1PN chirp is still a one-parameter family labelled by β . If f_0 is allowed to vary, then β alone is not enough; the 1PN correction depends on the dimensionless combination β/f_0 .

A slightly cleaner equal-mass parametrisation is obtained by defining

$$\lambda = \pi \frac{GM}{c^3} f_0 = \left(\frac{5\beta}{24\pi f_0} \right)^{3/5}. \quad (27)$$

Then

$$\frac{dx}{d(f_0 t)} = \frac{24\pi}{5} \lambda^{5/3} x^{11/3} \left[1 - \frac{487}{168} \lambda^{2/3} x^{2/3} \right], \quad x = \frac{f}{f_0}. \quad (28)$$

This form shows explicitly that $f(t) = f_0 x(f_0 t; \lambda)$ for equal masses.

The time-frequency relation can be integrated analytically:

$$t(f) = \frac{1}{\beta} \int_1^{f/f_0} \frac{x^{-11/3}}{1 - \epsilon x^{2/3}} dx. \quad (29)$$

With the substitution

$$u = x^{-2/3}, \quad (30)$$

this becomes

$$t(f) = \frac{3}{2\beta} \left[\frac{u^4}{4} + \frac{\epsilon u^3}{3} + \frac{\epsilon^2 u^2}{2} + \epsilon^3 u + \epsilon^4 \log |u - \epsilon| \right]_{u=(f/f_0)^{-2/3}}^{u=1}. \quad (31)$$

This gives an analytic implicit relation $t = t(f)$. In contrast to the 0PN case, solving explicitly for $f(t)$ does not produce a simple closed form like Eq. (14). In practice one would either numerically invert Eq. (24) or integrate the 1PN ODE directly.

The phase can also be written analytically as a function of frequency:

$$\phi(f) = \phi_0 + 2\pi \int_{f_0}^f f' \frac{dt}{df'} df'. \quad (32)$$

Again, this naturally gives $\phi(f)$ rather than a simple explicit $\phi(t)$.